Python Lab exercise- 3 (Lists and tuples)

1. Write a program to input a list and a number. Append the number to the list and print the updated list
2. Write a program that inputs a list, replicates it twice and then prints the list in ascending and descending order
3. Write a program to find the maximum and minimum element in the list with its index
4. Write a program to find the average of numbers in the list
5. Write a program to search for an element in a list
6. Write a program to find the count of a number in a given list of numbers
7. Write a program to check whether the maximum element of a list in the first half or the second half of the list
8. Write a program to calculate the odd numbers and even numbers in a list
9. Writ a program to input a list and 2 numbers m and n. Create another list from those list elements which are divisible by both m and n
10. Write a program to create a list with positive and negative numbers. Create 2 lists one with positive numbers and all the negative numbersfrom the given list . Print all the 3 numbers
11. Write a program to read a list of numbers. Create another list which removes all the duplicates and contain only the unique elements
12. Write a program that takes 2 lists of the same size and add their elements together to form a new list whose elements are sums of the corresponding elements of the 2 lists. Print all the 3 lists
13. Write a program to input names of n students and store them in a tuple. Input a name from the user and find if this student is present in the tuple or not
14. Write a program that creates a tuple storing first 9 terms of Fibonacci series
15. Write program to check if a tuple contains any duplicate elements